Edge-AI Driven Elastic Mesh PSO for Low-Altitude UAV IoT Networks in Zero-Infrastructure Environments

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***Abstract*—A low-altitude UAV swarm searching a disaster area without infrastructure must also carry each victim report back to the responders over its own radio links. SENTINEL treats the swarm network as an elastic mesh: it stretches and breaks while UAVs search, and is re-formed only when a confirmed report must reach the responder. Elastic-Mesh PSO (EM-PSO) plans waypoints with a Signal-Spring that penalises predicted link outage instead of distance; delivery is triggered by the information each UAV has confirmed; and weak thermal detections are corroborated from a second viewpoint. A local minimum-spanning-tree projection keeps the delivery path reliable for every feasible step. In about 13 000 simulated missions with a 2.4 GHz outage channel model, 3–50 UAVs, six failure settings and seven tuned baselines (30 seeds, paired tests), SENTINEL reports 23–29 % more victims within 60 s than the best baseline in urban rubble, and PX4 software-in-the-loop runs agree with the simulator. Keeping the mesh connected at all times lowers timely reporting.**

***Index Terms*—UAV swarm, edge AI, particle swarm optimization, search and rescue, intermittent connectivity, low-altitude IoT.**

I. Introduction

After an earthquake or a collapse, small multirotor UAVs with thermal cameras can cover rubble quickly, and compact detectors make it possible to find people on board instead of streaming video to a server [1], [2]. A detection is useful, however, only once the responders receive it, and without infrastructure the report travels over the UAVs' own short-range links, which buildings block, fading weakens and failures cut. A swarm that stays connected delivers every report at once but searches a small area; a swarm that explores freely finds more victims but leaves delivery to chance. We call this the *discovery–delivery trade-off* and measure it directly: the share of victims whose report reaches the responder within 60 s.

Our first version of SENTINEL addressed the problem with a quadratic spring that kept each UAV within a fixed 50 m of its nearest neighbour. Three things were wrong with that design. A fixed radius says nothing about a link behind rubble; keeping the nearest neighbour close does not keep the network connected (Lemma 1); and a 25 m test area is smaller than the radio range, so every method stays connected there and the methods barely differ (Section VI). This revision keeps the elastic-mesh idea but changes what it means. The mesh is free to stretch and break during search and is re-formed for delivery only. The contributions are:

- **EM-PSO with an outage-based Signal-Spring**, which penalises the predicted outage of a link under path loss, obstructions, shadowing and fading, and is active only while a report is being delivered.
- **Information-triggered delivery**: a UAV leaves the search when a victim it confirmed has not reached the responder within a hold time T_h.
- **Corroborative multi-view sensing**: a weak detection asks a neighbour for a lateral second view, and views are fused so that an occluded angle cannot veto a clear one.
- **A statistical evaluation** against seven tuned baselines with 3–50 UAVs, four channel profiles and six failure settings, with confidence intervals and paired tests, plus a PX4/Gazebo cross-check. Code, scenarios and results are released.

II. Related Work

Connectivity-maintenance controllers keep the communication graph connected at every instant, by following the gradient of its algebraic connectivity λ_2 [3], [4] or with edge potentials that grow without bound near the range limit [5]; local topology control such as the local minimum spanning tree (LMST) reduces the links that must be kept [6]. These methods tie the whole team to the network and assume a disk radio model. Intermittent-connectivity planning lets robots disconnect and meet on a schedule [7], data ferrying moves data between known sources and a sink [8], and age-of-information work optimises the freshness of collected data [9], [10]; all of them know the sources or the meetings in advance, whereas the victims are found by the search itself. Distributed PSO and Bayesian multi-UAV search [11]–[14] assume communication is available when needed, and multi-agent reinforcement learning, with MAPPO as a strong general method [15], has been applied to communication-aware UAV coordination [16]. SENTINEL combines online-discovered

TABLE I
CLOSEST APPROACHES (✓: ADDRESSED; ○: PARTLY; –: NOT ADDRESSED).

Approach	online sources	deadline	obstr./ fading	failures	free search
λ_2 control [3], [4]	–	–	–	○	–
Edge potentials [5]	–	–	–	–	–
Intermittent conn. [7]	–	○	–	–	✓
Ferrying / AoI [8], [10]	–	✓	○	–	✓
PSO / Bayesian search [12], [13]	✓	–	–	–	✓
MARL [15], [16]	○	○	○	○	✓
Our first version	✓	–	–	○	○
SENTINEL	✓	✓	✓	✓	✓

sources, a reporting deadline, an obstruction- and fading-aware channel and failure-aware evaluation (Table I).

III. SYSTEM MODEL

N UAVs fly at altitude h over an area with tall buildings (no-fly zones that also block radio links) and low rubble that hides victims. A responder radio (node 0) at the launch point is the only sink for reports. Every slot Δt each UAV broadcasts a status message with its position, its target ledger and acknowledgements.

Channel. The SINR of the broadcast from i at j is

$$\gamma_{ij} = \frac{P_t G^2\, 10^{-(\mathrm{PL}(d_{ij})+n_{ij}L_o+X_{ij})/10}\,|h_{ij}|^2}{\sigma^2 + \sum_{k\in\mathcal{I}_j}\bar{P}_{kj}}, \qquad (1)$$

with log-distance path loss $\mathrm{PL}(d) = \mathrm{PL}(d_0) + 10\eta\log_{10}(d/d_0)$, n_{ij} obstacles crossing the link with loss L_o each, shadowing $X_{ij} \sim \mathcal{N}(0,\sigma_X^2)$ correlated over displacement [17], unit-mean Rician fading $|h_{ij}|^2$ (Rayleigh once obstructed) and colliding transmitters $\mathcal{I}_j$. A packet is decoded iff $\gamma_{ij} \geq \gamma_{\mathrm{th}}$. The outage probability averaged over fading and shadowing is

$$P_{\mathrm{out}}(d,n) = \mathbb{E}_X\left[F_{|h|^2}\left(\gamma_{\mathrm{th}}/(\bar{\gamma}(d,n)\,10^{X/10})\right)\right], \qquad (2)$$

and the *reliable range* is $R_\varepsilon(n) = \max\{d : P_{\mathrm{out}}(d,n) \leq \varepsilon\}$. The link budget uses $f = 2.4\,\mathrm{GHz}$, $P_t = 20\,\mathrm{dBm}$, $G = 2\,\mathrm{dBi}$ per side, $B = 20\,\mathrm{MHz}$, noise figure 7 dB (noise floor -94 dBm) and $\gamma_{\mathrm{th}} = 5\,\mathrm{dB}$, which allows 113 dB of path loss before fading margins ($\mathrm{PL}(1\,\mathrm{m}) = 40\,\mathrm{dB}$). Profile parameters follow the ranges reported for low-altitude and street-level links [18], [19]. Table II gives the resulting ranges: in urban rubble $R_{0.05} = 91\,\mathrm{m}$ with a clear line of sight and 48 m behind one obstruction. The 50 m radius of our first version therefore matches a link through one layer of rubble, and a second obstruction or a harsher profile shrinks it further.

Information and failures. Each UAV keeps a ledger of world-frame victim entries; detections within the association gate are merged, and an entry is *confirmed* when its fused confidence $c_e \geq c_{\mathrm{hi}}$ and *tentative* when $c_{\mathrm{lo}} \leq c_e < c_{\mathrm{hi}}$. Ledgers spread by gossip with acknowledgements; an entry is *secured* once the responder holds it and *lost* if every UAV holding it fails first. UAVs fail by crashing (with their ledger), by radio loss or by a gradual transmit-power brownout, singly, simultaneously, in cascades or as a Poisson process.

TABLE II
CHANNEL PROFILES AND RELIABLE RANGES $R_{0.05}(n)$ FOR n OBSTRUCTIONS (σ_X, L_o IN DB).

Profile	η	σ_X	K	L_o	$R(0)$ [m]	$R(1)$ [m]
open	2	2	10 dB	6	2434	684
suburban	2.5	4	6 dB	10	339	95
urban rubble	3	6	3 dB	6	91	48
severe	3.5	8	Rayleigh	8	35	21

Objective. Let τ_k be the time victim k is first confirmed and T_k the time, from mission start, at which its report reaches the responder ($T_k = \infty$ if never). The team maximises

$$J = \frac{1}{|\mathcal{V}|}\sum_{k\in\mathcal{V}} e^{-T_k/\theta} \qquad (3)$$

under decentralised information. Staying connected makes delivery immediate but limits discovery; free search does the opposite.

IV. SENTINEL

Mode switching. Let $\mathcal{E}_i(t)$ be the confirmed, unsecured entries UAV i is responsible for (it created them, or they are older than $3T_h$). UAV i switches from EXPLORE to DELIVER when

$$\max_{e\in\mathcal{E}_i(t)} (t - \tau_e) \geq T_h \qquad (4)$$

and back when $\mathcal{E}_i(t) = \emptyset$. $T_h = 0$ delivers every confirmation at once; $T_h \to \infty$ never interrupts the search. Connection to the responder is known locally from a time stamp that the responder issues and every UAV relays.

EM-PSO. Each planning period the UAV picks a waypoint in its reachable disk $\mathcal{D}_i$ by maximising

$$F_i(x) = w_c\hat{G}_i(x) + w_d\sum_{e\in\mathcal{T}_i} c_e e^{-\frac{\|x-x_e\|^2}{2s^2}} + \frac{w_g(\|x_i-g_i\| - \|x-g_i\|) - w_t\|x-x_i\|}{vT_p} - S_i(x), \qquad (5)$$

where $\hat{G}_i$ is the expected number of new detections of a look at x on the UAV's search map, $\mathcal{T}_i$ the tentative entries, g_i a frontier goal and w_t a small travel cost. The Signal-Spring acts only while the UAV is tethered (delivering and connected, or relaying a delivery):

$$S_i(x) = \lambda_i\sum_{j\in\mathcal{S}_i}\left(\frac{[P_{\mathrm{out}}(\|x-x_j\|, n(x,x_j)) - \varepsilon_i]_+}{\varepsilon_i}\right)^2, \qquad (6)$$

with $\lambda_i = \lambda_0(1+\beta U_i)$, $\varepsilon_i = \varepsilon/(1+\beta U_i)$ and U_i the number of entries only i holds. It penalises predicted outage rather than distance, so it accounts for obstructions, and it tightens with the unique information a UAV carries. A constriction-coefficient PSO ($\chi = 0.7298$, $c_1 = c_2 = 2.05$) [11] maximises F_i.

Link preservation. $\mathcal{S}_i$ contains the neighbours adjacent to i in the minimum spanning tree of the reliable graph among the neighbours i has heard [6]. If all UAVs use the same positions

and have heard all their reliable neighbours, the mutually selected edges contain the global minimum spanning forest and so preserve the connected components.

Lemma 1. *Nearest-neighbour distance constraints do not imply connectivity.*

Proof. UAVs at $(0,0),(10,0),(200,0),(210,0)$ with $R = 50\,\mathrm{m}$ each have a neighbour at 10 m, yet no edge joins the two pairs. □

While tethered, UAV i scales its step u_i to the largest $\alpha \in [0,1]$ such that $x_i' = x_i + \alpha u_i$ satisfies, for all $j \in \mathcal{S}_i$,

$$\|x_i' - m_{ij}\| \le \tfrac{1}{2} r_{ij}(x_i') - \tfrac{1}{2} v\, a_{ij}, \quad r_{ij} = R_\varepsilon\big(n^{\delta_{ij}}(x_i', \hat{x}_j)\big), \tag{7}$$

where $\hat{x}_j$ is j's last reported position of age a_{ij}, $m_{ij} = (x_i + \hat{x}_j)/2$, $\delta_{ij} = v(\Delta t + a_{ij})$ and n^δ counts obstacles inflated by δ.

Proposition 1. *If $j \in \mathcal{S}_i$, $i \in \mathcal{S}_j$, both UAVs are tethered, speeds are bounded by v, positions are exchanged with staleness at most a, and both steps satisfy* (7)*, then link (i,j) is ε-reliable at $t + \Delta t$.*

Proof sketch. Each midpoint estimate is within $va/2$ of the true midpoint, so the triangle inequality and (7) give $\|x_i' - x_j'\| \le \frac{1}{2}(r_{ij} + r_{ji})$. The new segment lies within δ_{ij} of $[x_i', \hat{x}_j]$, so the inflated count bounds the true one, and since R_ε is non-increasing in n, $\|x_i' - x_j'\| \le R_\varepsilon(n(x_i', x_j'))$. □

The guarantee holds for feasible steps; if (7) is infeasible the UAV hovers, and we measure the resulting violations (Section VI).

Corroborative sensing. A tentative entry ($c_{\mathrm{lo}} \le c_e < c_{\mathrm{hi}}$, with $c_{\mathrm{lo}} = 0.40$, $c_{\mathrm{hi}} = 0.70$) asks the nearest neighbour for a view from $\pm 15\,\mathrm{m}$ perpendicular to the observer–target line, admitted only if the flank point keeps that neighbour's preserved links feasible. At $h = 30\,\mathrm{m}$ the flank views the victim 26.6° off nadir, enough to look past the edge of a covering slab, while the slant range grows by only 12 %, so the detector's range penalty stays small. The ledger keeps the best confidence $c_{e,v}$ per viewing sector v. Naive-Bayes fusion treats a weak view from a blocked side as evidence of absence; with many UAVs such views outvote a clear one. SENTINEL fuses corroboratively,

$$\operatorname{logit} c_e = \max_v \operatorname{logit} c_{e,v} + \sum_{v \ne v^\star} [\operatorname{logit} c_{e,v}]_+, \tag{8}$$

where $v^\star$ is the best view; other views can only add support.

Complexity. A planning step evaluates P particles for I iterations, each costing $O(k)$ outage look-ups for $k = |\mathcal{S}_i|$ preserved neighbours, and the local spanning tree costs $O(k_h^2 \log k_h)$ for the k_h neighbours heard; no term grows with N directly, unlike the swarm-wide eigenvector needed by λ_2 control. Because the hinge in (6) is squared, S_i is continuously differentiable and adds no kink to the fitness.

Algorithm 1 SENTINEL agent i (every slot Δt)

```
record look; add geolocated detections to the ledger
broadcast status (x_i, ledger, acks, stamp, t_tether); merge received ones
if planning slot then
    update mode with (4); request flanks for own tentative entries
    if DELIVER and not connected then
        waypoint ← x_0
    else
        waypoint ← arg max_{x∈D_i} F_i(x) by PSO        ▷ S_i≡0 unless tethered
    end if
end if
u_i ← step toward waypoint; if tethered, scale by α* from (7)
x_i ← x_i + u_i
```

TABLE III
DEFAULT PARAMETERS (FROM THE CODE AND THE TUNED CONFIGURATION).

Kinematics	$\Delta t = 0.5\,\mathrm{s}$, $v = 5\,\mathrm{m/s}$, $h = 30\,\mathrm{m}$, footprint 12 m
Planning	every 1 s, horizon $T_p = 4\,\mathrm{s}$, PSO $P = 24$, $I = 12$
Tuned	$T_h = 116\,\mathrm{s}$, $\lambda_0 = 1.20$, $\beta = 1.09$, $w_g = 0.066$
Spring	$\varepsilon = 0.05$, $w_c = 1$, $w_d = 1$, $w_t = 0.05$
Sensing	$c_{\mathrm{lo}} = 0.40$, $c_{\mathrm{hi}} = 0.70$, flank $\pm 15\,\mathrm{m}$, gate 5 m, 8 sectors
Detector	$c_{\max} = 0.92$, $\gamma = 1.5$, $r_{\mathrm{ref}} = 35\,\mathrm{m}$, $\sigma_c = 0.08$, false alarms 0.02/look

V. IMPLEMENTATION

The algorithms form one ROS-independent Python library, executed both by a fast multi-agent simulator and by ROS 2 Humble nodes that fly PX4 x500 quadrotors in Gazebo Harmonic through Micro XRCE-DDS (Fig. 2). Each UAV runs a flight commander, a thermal-vision node (YOLO26n with geolocation), the SENTINEL agent and a camera bridge; all inter-UAV traffic passes through a radio emulator that drops packets by sampled SINR. UAVs fly at $h = 30\,\mathrm{m}$ and $v = 5\,\mathrm{m/s}$ with a 12 m camera footprint; $T_h = 116\,\mathrm{s}$ and the other free parameters were tuned (Section VI); Table III lists the defaults. Worlds are generated from the same seeds as the fast simulator, with heated Rescue Randy mannequins lying in the open or partly under concrete slabs and warm debris that causes false alarms (Fig. 1).

VI. EVALUATION

Setup. Each run generates a new disaster area from its seed (no-fly buildings, rubble, warm debris, 25 victims of which one is buried) and lasts 600 s. Unless stated, the area is $400 \times 400\,\mathrm{m}$, $N = 8$, the channel is urban rubble and UAVs fail as a Poisson process at $0.01\,\mathrm{s}^{-1}$ per swarm. The detector in the loop is a parameterised model of confidence versus range and view-dependent occlusion with false alarms from warm debris. The metric is the share of all victims reported to the responder within 60 s of first confirmation. It replaces the network-retention and discovery rates of our first version, which were coarse for ten victims and one run, with a per-victim measure that counts only reports that actually arrive. Baselines: B0 constriction PSO; B1 our first version (50 m nearest-neighbour spring); B2 PSO with an outage penalty on

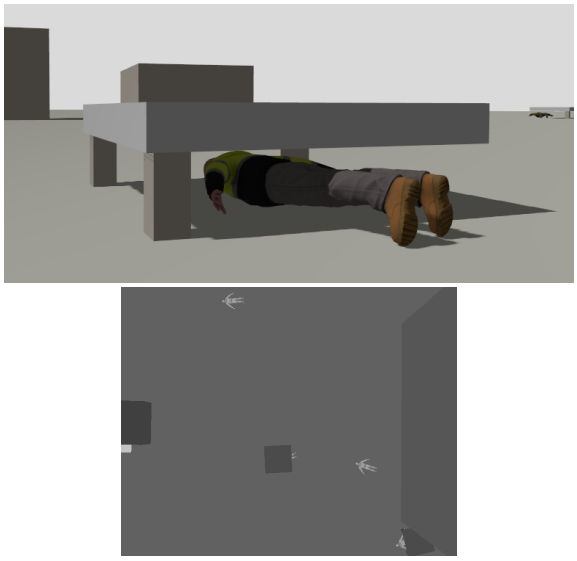


Fig. 1. Gazebo scene. Top: a victim partly under a slab (torso hidden from above). Bottom: white-hot thermal frame from a UAV at 30 m with victims in the open, under slabs and warm debris.

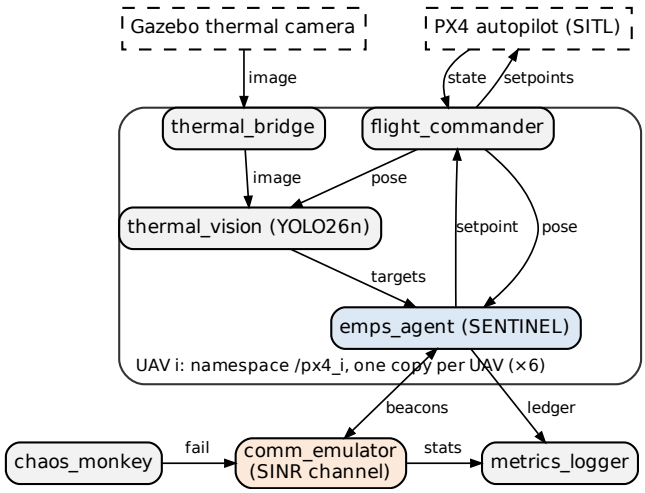


Fig. 2. ROS 2 node graph, transcribed from the running six-UAV system; one copy of the four UAV nodes runs per UAV, and UAVs exchange data only through the radio emulator.

all neighbours; B3 λ_2-gradient control [4] with the exact λ_2 (an oracle); B4 edge potentials with hysteresis [5]; B5 MAPPO [15]; B6 sweep search. Every method was tuned with the same Optuna budget on J using seeds that are never used for evaluation (MAPPO was trained to a plateau on separate seeds). Each setting uses seeds 0–29 with common random numbers; tables give the mean and the half-width of the 95 % bootstrap confidence interval, and * marks a paired Wilcoxon test against SENTINEL, Holm-corrected. The baselines keep the best single view per victim and request no second views, as in their designs; Table VII also gives them SENTINEL's fusion rule.

Channel and failures. SENTINEL reports the most victims

TABLE IV
VICTIMS REPORTED WITHIN 60 S PER CHANNEL PROFILE (400 M, $N = 8$, POISSON FAILURES). MEAN ± HALF-WIDTH OF THE 95 % BOOTSTRAP CI OVER 30 SEEDS; BEST IN BOLD; *: DIFFERS FROM SENTINEL (PAIRED WILCOXON, HOLM-CORRECTED, $p < 0.05$).

Method	open	suburban	urban	severe
SENTINEL	**0.58±0.04**	**0.56±0.05**	**0.57±0.05**	**0.43±0.05**
B0 PSO	0.47±0.05*	0.46±0.05*	0.45±0.04*	0.38±0.05
B1 v1 spring	0.46±0.05*	0.43±0.05*	0.41±0.05*	0.37±0.05
B2 comm-PSO	0.47±0.05*	0.37±0.04*	0.47±0.04*	0.34±0.05*
B3 λ_2-grad.	0.48±0.05*	0.45±0.05*	0.47±0.04*	0.35±0.06
B4 potential	0.50±0.05*	0.10±0.03*	0.13±0.04*	0.30±0.06*
B5 MAPPO	0.41±0.05*	0.42±0.04*	0.40±0.05*	0.33±0.05*
B6 sweep	0.42±0.05*	0.42±0.05*	0.42±0.05*	0.37±0.05

in every channel profile (Table IV), significantly more than all baselines except in the severe profile, where it leads (0.433 vs. 0.377, B0) without significance. Under all six failure settings in urban rubble it reports 23–29 % more victims than the best baseline (Table V). Losing a confirmed victim to failures is rare for every method because ledgers spread by gossip: at most 0.9 % averaged over the failure settings, and 8.5 % in the worst cell (twice the Poisson rate, severe channel; SENTINEL 3.2 %), so we claim no advantage in information loss.

Swarm size. On the 400 m area SENTINEL is significantly better than every baseline from 3 to 50 UAVs (Table VI). On a 200 m area, where contacts are frequent, it has the highest mean at every size but is significantly better than all baselines only for $N = 50$ (0.809 vs. 0.779 for MAPPO). Planning takes 4–11 ms per agent, well below the 1 s planning period.

What produces the gains. Table VII removes components on the same seeds. Removing fusion (0.519) or delivery (0.523) lowers the urban share significantly from 0.575; keeping the mesh connected at all times collapses it to 0.03–0.13 across regimes. Given SENTINEL's fusion rule, the best urban baseline rises to 0.507 (B3) and stays significantly below SENTINEL, as do all seven. With fusion, flank and delivery all removed SENTINEL still reports 0.524, above every baseline with fusion; the remaining differences from the baselines are tuned exploration aids and responder re-attachment, which we do not claim as contributions. In a dense 50-UAV swarm the picture changes: fusion is decisive for SENTINEL (0.809 vs. 0.740 without it and 0.656 with naive-Bayes fusion), but baselines given the same rule reach 0.815–0.865, and B4 exceeds SENTINEL significantly. The delivery trigger helps with eight UAVs (Fig. 3): $T_h = 15$ s raises the share over never delivering from 0.523 to 0.599 in urban rubble and from 0.419 to 0.501 in the severe channel, where the 90th-percentile latency falls from 74 s to 41 s; with four UAVs, where contacts are rare, the differences are not significant. The tuned $T_h = 116$ s optimises J, not the 60 s deadline.

Link preservation. Over all experiments, tethered and mutually preserved links that were reliable before a step failed after it 65 times in 1 317 947 link-steps (4.9×10^{-5}). On the same seeds without the projection, violations rose from 1 in 1 663 link-steps (1 run) to 42 in 1 868 (21 runs), a 37-fold

TABLE V

VICTIMS REPORTED WITHIN 60 S UNDER SIX FAILURE SETTINGS (URBAN RUBBLE, 400 M, $N = 8$). MEAN ± HALF-WIDTH OF THE 95 % BOOTSTRAP CI OVER 30 SEEDS; BEST IN BOLD; *: DIFFERS FROM SENTINEL (PAIRED WILCOXON, HOLM-CORRECTED, $p < 0.05$).

Method	none	single crash	30 % simult.	cascading	Poisson×2	brownout
SENTINEL	**0.69±0.04**	**0.69±0.04**	**0.67±0.04**	**0.64±0.04**	**0.46±0.05**	**0.69±0.04**
B0 PSO	0.54±0.04*	0.52±0.04*	0.51±0.04*	0.51±0.04*	0.34±0.06*	0.54±0.04*
B1 v1 spring	0.50±0.05*	0.50±0.05*	0.50±0.05*	0.49±0.05*	0.36±0.05*	0.51±0.05*
B2 comm-PSO	0.56±0.03*	0.55±0.03*	0.52±0.03*	0.52±0.04*	0.34±0.03*	0.55±0.04*
B3 λ_2-grad.	0.53±0.03*	0.52±0.04*	0.51±0.04*	0.49±0.03*	0.34±0.05*	0.53±0.03*
B4 potential	0.19±0.04*	0.16±0.03*	0.14±0.04*	0.16±0.04*	0.09±0.03*	0.18±0.04*
B5 MAPPO	0.50±0.04*	0.47±0.04*	0.45±0.04*	0.46±0.04*	0.33±0.05*	0.49±0.04*
B6 sweep	0.52±0.04*	0.51±0.04*	0.49±0.04*	0.49±0.04*	0.35±0.05*	0.52±0.04*

TABLE VI

VICTIMS REPORTED WITHIN 60 S VS. SWARM SIZE (400 M, URBAN RUBBLE, POISSON FAILURES). MEAN ± HALF-WIDTH OF THE 95 % BOOTSTRAP CI OVER 30 SEEDS; BEST IN BOLD; *: DIFFERS FROM SENTINEL (PAIRED WILCOXON, HOLM-CORRECTED, $p < 0.05$).

Method	$N = 3$	$N = 5$	$N = 10$	$N = 20$	$N = 50$
SENTINEL	**0.20±0.04**	**0.39±0.06**	**0.67±0.04**	**0.75±0.03**	**0.79±0.03**
B0 PSO	0.15±0.03*	0.29±0.05*	0.53±0.04*	0.62±0.03*	0.66±0.04*
B1 v1 spring	0.11±0.03*	0.23±0.05*	0.50±0.05*	0.65±0.03*	0.71±0.04*
B2 comm-PSO	0.15±0.04*	0.31±0.05*	0.53±0.04*	0.64±0.04*	0.69±0.03*
B3 λ_2-grad.	0.14±0.04*	0.27±0.05*	0.53±0.04*	0.64±0.04*	0.67±0.03*
B4 potential	0.02±0.01*	0.03±0.02*	0.24±0.05*	0.49±0.05*	0.71±0.04*
B5 MAPPO	0.15±0.04*	0.27±0.04*	0.50±0.05*	0.63±0.03*	0.73±0.03*
B6 sweep	0.15±0.04*	0.27±0.05*	0.44±0.04*	0.49±0.04*	0.62±0.03*

TABLE VII

WHAT PRODUCES THE GAINS: COMPONENTS REMOVED (SAME SEEDS) AND THE BEST BASELINE WITH AND WITHOUT SENTINEL'S FUSION RULE; VICTIMS REPORTED WITHIN 60 S. DENSE: 200 M. MEAN ± HALF-WIDTH OF THE 95 % BOOTSTRAP CI OVER 30 SEEDS; BEST IN BOLD; *: DIFFERS FROM SENTINEL (PAIRED WILCOXON, HOLM-CORRECTED, $p < 0.05$).

Variant	urban, N=8	severe, N=8	dense, N=50
SENTINEL	**0.57±0.05**	**0.43±0.05**	0.81±0.03
no fusion	0.52±0.05*	0.41±0.04	0.74±0.04*
no delivery (T_h=∞)	0.52±0.05*	0.42±0.05	–
no fusion, flank, delivery	0.52±0.05*	0.40±0.06	–
always connected	0.13±0.03*	0.06±0.02*	–
best baseline	0.47±0.04* B3	0.38±0.05 B0	0.78±0.04* B5
best baseline + fusion	0.51±0.04* B3	0.42±0.06 B1	**0.87±0.03*** B4

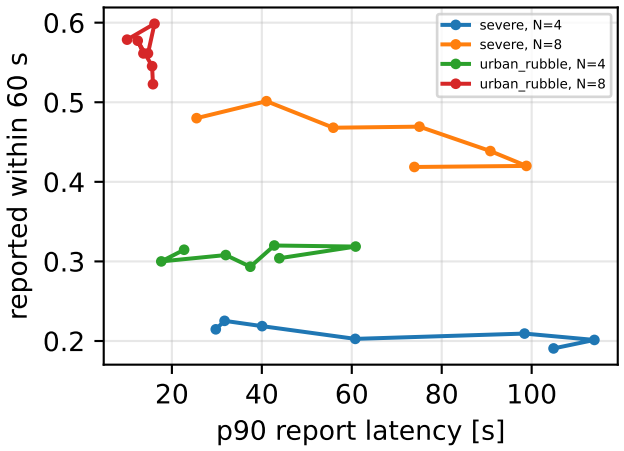


Fig. 3. Hold time $T_h \in \{0, 15, 30, 60, 120, 240, \infty\}$ s: share reported within 60 s vs. 90th-percentile latency (mean of 30 seeds per point).

difference (paired Wilcoxon, $p < 10^{-4}$). Tethers are short, so this protection does not change the on-time share measurably.

PX4/Gazebo and the earlier scenario. 3 PX4 x500 UAVs flew SENTINEL and B0 in Gazebo on 3 generated 100 m areas with one crash; reported victims, packet delivery ratio and connected time agree with the fast simulator within 0.07 (SENTINEL 0.53 vs. 0.60, B0 0.60 vs. 0.62). With 3 seeds this is a consistency check, and in an area this size B0 is slightly ahead in both environments. Re-running the 25 m, 3-UAV scenario of our first version, every method stays connected throughout. That scenario has no responder, so we compare confirmed victims: SENTINEL confirms 0.71 of them, B0 0.67 (not significantly fewer) and the first version 0.63 (significantly fewer); victims closer than the 5 m association gate merge in 7–10 % of confirmations. The network-retention gains reported for that scenario are not reproduced, and we withdraw them.

Detector. Trained under one protocol on HIT-UAV [1], YOLO26n [2] matches YOLOv8n in validation mAP50 (0.948 vs. 0.947; mAP50-95 0.533 vs. 0.520) with 19 % fewer parameters and 31 % fewer operations. End-to-end TensorRT FP16 latency on a desktop RTX 3060 is 1.81 ms vs. 1.73 ms; embedded latency was not measured.

VII. LIMITATIONS AND CONCLUSION

All results come from simulation, a fast multi-agent simulator and PX4 software-in-the-loop; no hardware flights were made. The channel is parameterised from published ranges and not calibrated in rubble. Detection accuracy is the training run's validation result on HIT-UAV, whose imagery is taken from higher altitudes than our 30 m; the detector transfers poorly to rendered scenes, so the loop uses a parameterised detector model whose parameters were set by hand and swept, not fitted. Victims closer than the association gate can merge. MAPPO was trained with one observation and reward design. Latency was measured on a desktop GPU only.

SENTINEL treats the swarm network as an elastic mesh that stretches during search and is re-formed for delivery. Across 13 000 simulated missions it reported more victims within 60 s than seven tuned baselines in most settings and was never

significantly worse than any of them. Fusion and information-triggered delivery had significant effects, although in dense swarms the fusion rule alone explains the advantage, and constant connectivity reduced timely reporting. Future work will adapt T_h to the observed contact rate, calibrate the channel with measurements and fly the system on hardware.

REFERENCES

[1] J. Suo, T. Wang, X. Zhang, H. Chen, W. Zhou, and W. Shi, "HIT-UAV: A high-altitude infrared thermal dataset for unmanned aerial vehicle-based object detection," *Scientific Data*, vol. 10, p. 227, 2023.

[2] Ultralytics, "Ultralytics YOLO26," Software, ultralytics v8.4.0 (released 14 Jan. 2026), https://docs.ultralytics.com/models/yolo26/, 2026.

[3] M. M. Zavlanos and G. J. Pappas, "Distributed connectivity control of mobile networks," *IEEE Transactions on Robotics*, vol. 24, no. 6, pp. 1416–1428, 2008.

[4] L. Sabattini, N. Chopra, and C. Secchi, "Decentralized connectivity maintenance for cooperative control of mobile robotic systems," *The International Journal of Robotics Research*, vol. 32, no. 12, pp. 1411–1423, 2013.

[5] M. Ji and M. Egerstedt, "Distributed coordination control of multiagent systems while preserving connectedness," *IEEE Transactions on Robotics*, vol. 23, no. 4, pp. 693–703, 2007.

[6] N. Li, J. C. Hou, and L. Sha, "Design and analysis of an MST-based topology control algorithm," in *Proc. IEEE INFOCOM*, vol. 3, 2003, pp. 1702–1712.

[7] Y. Kantaros and M. M. Zavlanos, "Distributed intermittent connectivity control of mobile robot networks," *IEEE Transactions on Automatic Control*, vol. 62, no. 7, pp. 3109–3121, 2017.

[8] K. Cui, L. Baumgärtner, M. B. Yilmaz, M. Li, C. Fabian, B. Becker, L. Xiang, M. Bauer, and H. Koeppl, "UAV swarms for joint data ferrying and dynamic cell coverage via optimal transport descent and quadratic assignment," in *Proc. 48th IEEE Conference on Local Computer Networks (LCN)*, 2023, pp. 1–8.

[9] S. Kaul, R. Yates, and M. Gruteser, "Real-time status: How often should one update?" in *Proc. IEEE INFOCOM*, 2012, pp. 2731–2735.

[10] M. Bafarassat and S. Coleri, "Reliability- and connectivity-constrained age-of-information optimization for UAV swarm IoT data collection," *IEEE Internet of Things Journal*, 2026, early access; arXiv:2608.00061.

[11] M. Clerc and J. Kennedy, "The particle swarm—explosion, stability, and convergence in a multidimensional complex space," *IEEE Transactions on Evolutionary Computation*, vol. 6, no. 1, pp. 58–73, 2002.

[12] L. Yang, Y. Hao, J. Xu, and M. Li, "Multi-UAV collaborative target search method in unknown dynamic environment," *Sensors*, vol. 24, no. 23, p. 7639, 2024.

[13] D. Niu, P. Yi, and Y. Hong, "Distributed event-driven Bayesian search for multi-UAV systems with spatially correlated targets," *Sensors*, vol. 26, no. 16, p. 5189, 2026.

[14] J. Zhu, X. Zhou, Z. Zi, Y. Xie, X. Jiang, and Z. Zhang, "Cooperative search method of multi-UAVs for mountain search and rescue missions," *Drones*, vol. 10, no. 7, p. 493, 2026.

[15] C. Yu, A. Velu, E. Vinitsky, J. Gao, Y. Wang, A. Bayen, and Y. Wu, "The surprising effectiveness of PPO in cooperative multi-agent games," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 35, 2022, pp. 24 611–24 624.

[16] R. Shi, X. Yu, Y. Wang, Y. Tian, Z. Liu, W. Wu, X.-P. Zhang, and M. M. Veloso, "Symmetry-informed MARL: A decentralized and cooperative UAV swarm control approach for communication coverage," *IEEE Transactions on Mobile Computing*, vol. 24, no. 9, pp. 8039–8056, 2025.

[17] M. Gudmundson, "Correlation model for shadow fading in mobile radio systems," *Electronics Letters*, vol. 27, no. 23, pp. 2145–2146, 1991.

[18] 3GPP, "Enhanced LTE support for aerial vehicles," 3rd Generation Partnership Project, Technical Report TR 36.777, V15.0.0, 2017.

[19] ITU-R, "Propagation data and prediction methods for the planning of short-range outdoor radiocommunication systems and radio local area networks," Recommendation ITU-R P.1411-12, Aug. 2023.